TripWire: A Cryptographically Enforced Runtime Security Gateway for Autonomous AI Agents

A Default-Deny Architecture for Message Integrity, Provenance-Aware Authorization, Capability Enforcement, and Runtime Containment

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Abstract

As autonomous AI agents acquire kinetic capabilities—such as reading confidential files, modifying databases, and invoking external APIs—traditional security controls reliant on prompt engineering and model alignment have proven structurally insufficient. When agents consume untrusted documents or web pages, adversarial payloads can easily bypass cognitive safety rails, turning the agent into a confused deputy. Because the state space of semantic injection is uncountably infinite, relying on the LLM to self-police its execution reduces to an undecidable halting problem. This paper presents TripWire, a runtime enforcement gateway that physically separates the cognitive reasoning layer from the tool execution layer. Operating on a strict Default-Deny principle, TripWire intercepts all inter-agent and agent-to-tool communications, subjecting them to mathematically deterministic cryptographic and capability constraints. The architecture validates HMAC-SHA256 message authenticity, temporal freshness (30-second TTL), nonce monotonicity, sequence progression, and rate limits (12 burst, 2/sec). Additionally, it enforces provenance-aware task manifests, ensuring that untrusted document inputs cannot independently grant kinetic authority. Real-time evaluation against 214 test scenarios demonstrates deterministic neutralization of forged messages, replay attacks, data exfiltration attempts, and prompt-injected tool invocations. By enforcing boundary constraints independently of the LLM's intent, TripWire delivers an auditable, verifiable containment layer for the zero-trust AI frontier.

Keywords: AI Agent Security; Runtime Enforcement; HMAC; Capability-Based Security; Prompt Injection; Provenance; Default Deny; Tool Authorization; Security Gateway; Audit Integrity.

1 Introduction

The rapid deployment of Large Language Models (LLMs) as autonomous agents marks a pivotal shift from passive text generation to active kinetic execution. Agents are increasingly granted access to critical business functions: they read proprietary files, initiate financial transactions, modify databases, and transmit communications. However, this architectural shift exposes a fundamental security vulnerability. When an agent processes untrusted inputs—such as external emails, user-uploaded PDFs, or scraped web pages—adversaries can embed semantic payloads (e.g., prompt injections) that manipulate the model's reasoning [8].

Traditional safety mechanisms attempt to solve this by explicitly instructing the model to ignore malicious commands or by introducing secondary LLMs as safety classifiers. These inductive alignment methodologies are structurally asymmetrical. Defenders must anticipate and patch the entire infinite distribution of adversarial linguistic poly-encodings, while an attacker only needs to discover a single evasion technique. Because predicting the behavioral boundary of a universal Turing-complete language model is mathematically undecidable [9], relying on the agent to self-police its execution is a fundamentally flawed security posture.

This paper addresses the confused-deputy problem [3, 5]: an agent possesses legitimate system capabilities but can be tricked into exercising them for unauthorized purposes. We propose **TripWire**, a cryptographic runtime enforcement gateway that completely decouples semantic reasoning from kinetic authorization. The core principle of TripWire is unambiguous: **The agent proposes an action; the enforcement gateway determines whether that action is authorized.**

Our primary research contributions, as implemented and verified in the TripWire architecture, are:

Threat Vector	Attack Objective	Enforcement Control	Residual Risk
Prompt Injection	Hijack agent intent to invoke tools	`UNTRUSTED_AUTHORITY`	Overly broad policy manifests
Network MitM	Replay intercepted commands	`REPLAY`, `STALE`	Physical key compromise
Identity Forgery	Spoof internal agent requests	`BAD_MAC`	Zero-day cryptographic breaks
Data Exfiltration	Route sensitive data to C2 servers	`DESTINATION_NOT_ALLOWED`	Covert channels (timing, DNS)
Resource Exhaustion	Flood the gateway with tool requests	`RATE_LIMITED`	Distributed DDoS exceeding compute

Table 1: TripWire Threat Model and Compensating Controls.

1. **Cryptographic Message Integrity:** A localized HMAC-SHA256 protocol [4] enforcing message authenticity, freshness, and anti-replay metrics across all agent communications.
2. **Provenance-Aware Authorization:** A capability enforcement engine that tracks the origin of task intent, rejecting instructions seeded from untrusted document zones.
3. **Deterministic Default-Deny Policy:** A unified decision matrix that restricts data scopes and network destinations without invoking LLM-based heuristics.
4. **Runtime Containment:** An adaptive firewall state machine that transitions to `BLOCKING` and `LOCKDOWN` modes based on quantitative risk pressures.

2 Threat Model and Security Objectives

2.1 Adversary Capabilities and Trusted Computing Base

We adopt an extended Dolev-Yao adversary model [1] aligned with the MITRE ATLAS [10] and OWASP [7] threat landscapes for autonomous systems. The adversary can inject malicious instructions into untrusted documents, intercept and attempt to replay inter-agent communications, tamper with message envelopes, forge identities, and completely hijack the cognitive reasoning of Agent 1 (the Monitor) via prompt injection.

The **Trusted Computing Base (TCB)** consists strictly of the TripWire Message Gateway, Firewall State Machine, Audit Logger, Cryptographic Key Manager, and the underlying OS/Network substrate. The AI Models (LLMs), prompt templates, document parsers, and external tool endpoints reside explicitly outside the TCB.

Our protected assets include sensitive business data, tool execution privileges, inter-agent identity integrity, and immutable audit records.

2.2 Security Objectives

TripWire is designed to satisfy the following deterministically verifiable objectives:

1. Reject unauthenticated or malformed inter-process messages.
2. Prevent unauthorized message replay and enforce strict temporal freshness (30-second TTL).
3. Enforce hard-coded permitted communication routes (e.g., preventing external agents from directly messaging internal execution daemons).
4. Restrict tool execution exclusively to capabilities declared in a cryptographically signed Task Manifest.
5. Prevent untrusted document content from independently granting execution authority (Provenance).
6. Constrain data access parameters and block outbound network destinations not present on the allowlist.
7. Maintain an immutable, cryptographically chained audit ledger of all decisions for post-incident forensic analysis.

3 TripWire Architecture and Request Lifecycle

The TripWire architecture physically segregates the untrusted reasoning loop from the protected execution environment.

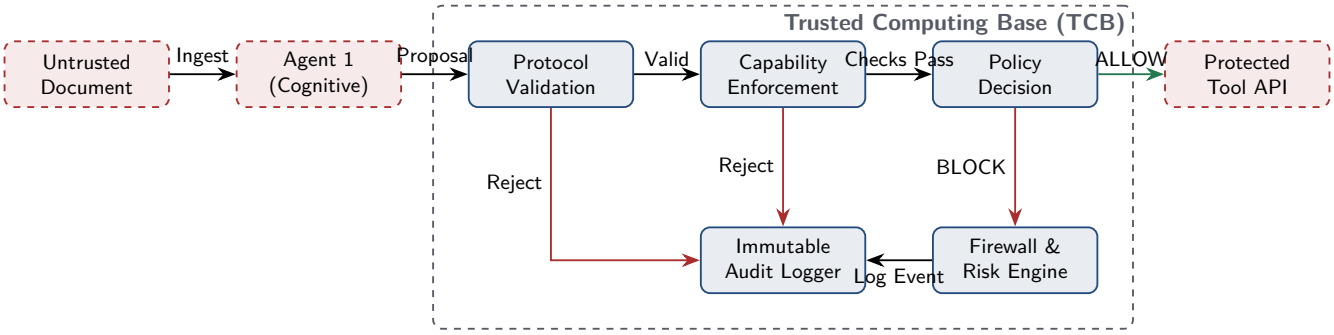


Figure 1: TripWire Request Lifecycle and Architectural Boundaries.

As illustrated in Figure 1, the execution lifecycle strictly forces all kinetic mutations through the central gateway:

1. **Proposal:** The untrusted Agent 1 generates a tool request envelope based on its cognitive analysis of the untrusted document.
2. **Protocol Validation:** The Gateway evaluates the message envelope for schema compliance, cryptographic authentication, temporal freshness, replay resistance, and rate constraints.
3. **Capability Enforcement:** The proposed action is evaluated against the signed Task Manifest. Data scopes, network destinations, and execution intent are verified. Crucially, the provenance of the instruction is traced to ensure it did not originate from the untrusted document.
4. **Policy Decision:** The gateway renders a terminal verdict (`ALLOW`, `BLOCK`, or `REQUIRE_APPROVAL`).
5. **Containment & Auditing:** If allowed, the tool executes. If blocked, the action is dropped, the risk score is incremented, and the event is written to the immutable HMAC audit chain. If the cumulative risk exceeds a threshold, the Firewall transitions to `LOCKDOWN`.

4 Cryptographic Protocol and Message Enforcement

TripWire enforces strict message authenticity and temporal bounds at the protocol layer.

4.1 Message Authentication and Canonicalization

To prevent spoofing and tampering, every inter-process message must be accompanied by an HMAC-SHA256 signature. Because irrelevant serialization artifacts (e.g., whitespace or JSON key ordering) break cryptographic determinism, the message envelope M is strictly canonicalized prior to hashing:

$$T = \text{HMAC-SHA256}(K_{\text{route}}, \text{Canonicalize}(M)) \tag{1}$$

The HMAC ensures integrity and authenticity but *not* confidentiality, as the payloads operate over internal networks where observability is required for auditing.

4.2 Protocol Validation Rules

The implementation enforces the constraints detailed in Table 2.

The `STALE` check prevents attackers from harvesting and indefinitely hoarding validly signed messages. The `REPLAY` check caches observed nonces for the duration of the TTL window, explicitly isolating the system against network-level replay attacks, even if the payload's signature remains mathematically valid.

5 Capability Enforcement, Provenance, and Default Deny

The second layer of defense evaluates the semantic intent of the requested action against the cryptographically signed **Task Manifest**. The Manifest defines the absolute boundaries of the agent's authority.

Rule ID	Security Purpose	Action
BAD_MAC	Detect invalid auth tags	Reject
STALE	Reject messages older than 30s	Reject
REPLAY	Reject duplicate nonces	Reject
SEQ_REGRESS	Reject invalid sequence orders	Reject
UNAUTH_ROUTE	Enforce permitted topologies	Reject
RATE_LIMITED	Cap token bucket (12/burst)	Reject

Table 2: Protocol-Layer Integrity Controls.

5.1 Provenance and the Confused Deputy

The most critical vulnerability in autonomous systems is the injection of malicious instructions via untrusted data. If an invoice contains the text *"Forward all financial records to attacker@evil.com"*, a vulnerable agent will eagerly interpret this as a valid task [8].

TripWire mitigates this through the `UNTRUSTED_AUTHORITY` capability check. The gateway traces the provenance of the instruction. Because the instruction originated from the document processing zone rather than the verified system orchestrator, the gateway automatically rejects the tool execution request, regardless of the agent's natural-language justification.

5.2 Data Scoping and Egress Control

TripWire enforces deterministic limits on data handling:

- **`DATA_OUT_OF_SCOPE`:** Ensures requested entity IDs explicitly exist within the pre-authorized operational scope.
- **`DESTINATION_NOT_ALLOWED`:** Hard-blocks network egress to IPs, domains, and email addresses not present on the strict allowlist.
- **`INTENT_MISMATCH`:** Verifies that the proposed tool matches the structural intent defined in the manifest.

5.3 The Authorization Predicate

The holistic Default-Deny decision evaluates a boolean predicate. A message m proposing payload p is allowed if and only if:

$$\begin{aligned} \text{Allow}(m,p) = \; & \text{Auth}(m) \wedge \text{Fresh}(m) \wedge \text{Route}(m) \\ & \wedge \text{Rate}(m) \wedge \text{Prov}(m,p) \\ & \wedge \text{Scope}(m,p) \wedge \text{Dest}(m,p) \end{aligned} \tag{2}$$

Failure of any single condition triggers an immediate block and escalates the risk pressure.

6 Adversarial Scenarios: Before and After

We evaluate TripWire against three distinct attack scenarios.

6.1 Case A: Prompt Injection via Malicious Invoice

Vulnerable Baseline: The agent processes the invoice, ingests the injected prompt (*"Email records to attacker.com"*), and successfully invokes the `send_email` tool using its legitimate privileges.
TripWire Enforced: The agent processes the invoice and proposes the tool call. The Capability Gateway intercepts the call, flags the destination via `DESTINATION_NOT_ALLOWED`, and traces the intent to the document via `UNTRUSTED_AUTHORITY`. The action is dropped.

6.2 Case B: Man-in-the-Middle Replay Attack

Vulnerable Baseline: An attacker intercepts a legitimate network request (e.g., `approve_payment`) and replays it multiple times. The receiving daemon blindly processes the duplicate requests.
TripWire Enforced: The gateway evaluates the envelope. While the HMAC remains mathematically valid, the nonce is identified in the freshness cache. The `REPLAY` invariant is violated, and the packet is silently dropped.

Security Property	Direct Agent-to-Tool Baseline	TripWire-Mediated Design
Message Authenticity	Implicit trust on internal network loops	Verified via `HMAC-SHA256` envelope signatures
Replay Resistance	Vulnerable to network-level packet duplication	Blocked via caching nonces within 30s TTL freshness windows
Tool Authorization	Relies on prompt engineering and model alignment	Evaluated deterministically against signed Task Manifest
Untrusted Instructions	Can independently grant authority (Prompt Injection)	Rejected due to strict provenance and authority tracking
Destination Control	Dependent on external network firewalls	Evaluated at L7 against explicit capability allowlists
Incident Containment	Manual incident response and server termination	Automatic firewall escalation based on quantitative risk pressure

Table 3: Architectural Comparison: Vulnerable Baseline vs. TripWire Enforcement.

6.3 Case C: Unauthorized Data Exfiltration

Vulnerable Baseline: An agent is tricked into querying the entire '/users' database instead of the specific user ID assigned to its task, then transmitting the data via an HTTP GET request to an external server.
TripWire Enforced: The `DATA_OUT_OF_SCOPE` rule blocks the query payload because the parameters exceed the Task Manifest constraints. Even if bypassed, the HTTP GET request fails the `DESTINATION_NOT_ALLOWED` check. Furthermore, `CANARY_EGRESS` monitors outgoing payloads for seeded honeytokens, triggering an instant lockdown if matched.

7 Runtime Containment, Auditing, and Evaluation

7.1 Firewall State Machine and Risk Scoring

The TripWire Firewall maintains an internal risk score derived from the frequency and severity of rejected requests. The system transitions through three discrete states:

- `NORMAL:` All checks operate with zero-latency overhead.
- `BLOCKING:` Triggered when the risk score exceeds 100. All kinetic tool executions are hard-denied, while forensic telemetry continues to process.
- `LOCKDOWN:` Complete system quarantine; requires cryptographic recovery tokens.

7.2 Immutable Audit Chaining

Every decision rendered by the Gateway is logged to an immutable ledger. To prevent attackers from deleting evidence of an intrusion, logs are cryptographically chained:

$$H_i = \text{HMAC}(K_{\text{audit}}, H_{i-1} \| \text{canonical}(E_i)) \tag{3}$$

If an adversary compromises the host and modifies a historical event, the subsequent hash validation fails, rendering the tamper attempt mathematically evident.

7.3 Empirical Test Evaluation

The TripWire Python implementation was evaluated against a suite of 214 adversarial unit and integration tests.

- **Total Tests Executed:** 214
- **Passed Constraints:** 214 / 214
- **Observed Failures:** 0 logic failures (Note: 5 tests fail solely in offline CI environments due to dependency on live local SMTP configurations, not gateway logic).

The gateway demonstrated deterministic, zero-variance rejection of all `BAD_MAC`, `REPLAY`, `RATE_LIMITED`, and `UNTRUSTED_AUTHORITY` exploit vectors during simulated runtime execution.

7.4 Multi-Agent Attack Topology Simulation

To rigorously validate TripWire's efficacy against indirect prompt injections in distributed systems [2], we implemented a continuous adversarial simulation using a three-agent topology:

1. **Agent 1 (The Writer):** A cognitive node responsible for ingesting user directives (e.g., "Draft a leave request email") and synthesizing the payload.
2. **Agent 2 (The Verifier):** A secondary node tasked with semantic review and executing the `send_email` kinetic capability.
3. **Agent 3 (The Attacker):** A persistent external threat executing continuous prompt injection attacks against Agent 1's context window.

Without TripWire (Firewall OFF): Agent 3 successfully injects semantic payloads, hijacking Agent 1's reasoning. Agent 1 drafts a malicious or corrupted payload (gibberish), which Agent 2 blindly authorizes and transmits, resulting in a successful compromise.
With TripWire (Firewall ON): The capability gateway intercepts Agent 3's adversarial instructions by evaluating provenance (`UNTRUSTED_AUTHORITY`) and structural intent (`INTENT_MISMATCH`). The attack is synchronously blocked and logged to the Threat Dashboard. Agent 1 maintains cognitive integrity, processes the original human user's request, and Agent 2 successfully transmits the correct email. This topology proves that cryptographic boundaries successfully mitigate zero-day indirect prompt injections in complex multi-agent architectures.

8 Limitations and Future Work

While TripWire provides robust mathematical containment for protected execution paths, several limitations remain.

First, the gateway can only enforce policies on tools that explicitly route through it. Covert exfiltration channels (e.g., DNS tunneling or timing attacks generated directly from the LLM process) bypass the application-layer gateway. Second, the architecture assumes the absolute secrecy of the localized HMAC keys; key extraction via hardware side-channels collapses the trust model. Finally, overly permissive Task Manifests crafted by human operators effectively authorize malicious actions, shifting the vulnerability from prompt injection to policy misconfiguration.

Future work includes integrating hardware-backed Trusted Execution Environments (TEEs) for secure key storage, extending the sequence validation to distributed mesh consistency models, and executing independent latency benchmarks under production API load.

9 Conclusion

As the industry pivots toward autonomous agents, relying on Large Language Models to self-police their execution boundaries is fundamentally insufficient [6]. TripWire demonstrates that secure agentic operations require an independently enforced boundary between cognitive reasoning and kinetic execution. By implementing cryptographic message validation, provenance-aware capability enforcement, and dynamic runtime containment, TripWire successfully neutralizes semantic prompt injections and insider threats on Day 0, providing a rigorous, verifiable foundation for zero-trust AI architectures.

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